Unconditional Cash Transfers and Youth Labor Supply: Evidence from South Korea *

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Abstract

This paper studies how unconditional income support affects early-career labor-market decisions using the Seongnam Youth Dividend, a universal cash transfer introduced in 2016 in South Korea. Seongnam-si, a municipality in Korea, provided all 24-year-old residents with one million KRW, paid in four quarterly installments over one year. Using population-wide administrative data from Statistics Korea and a stacked difference-in-differences design comparing Seongnam with neighboring cities, we estimate the effects up to three years after payments end. The transfer temporarily reduces labor supply during the payment period: employment declines by about 1 percentage point from a 44 percent baseline, driven largely by reductions in regular employment. After the transfer ends, employment and earnings gradually recover toward baseline, with little evidence of persistent decline. The employment decline is concentrated among individuals with no positive labor income during the year preceding eligibility, with further evidence pointing to delayed entry. Among previously non-employed individuals who enter employment after eligibility, treated entrants are more likely to enter mid-sized firms and less likely to enter microenterprises than control entrants. These patterns suggest that the delayed entry may reflect an extended search for more favorable employment opportunities.

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1 Introduction

Young adults often face a trade-off between earning income immediately and investing in activities that improve longer-run labor market prospects, such as job search, training, or further education. Income support for young adults is motivated by the idea that easing short-term financial pressure allows more productive early-career choices. In recent years, unconditional cash transfer (UCT) and universal basic income (UBI) pilots in Finland, Spain, Canada, and the United States have tested whether such support changes labor supply (see Appendix Table A.1).

Whether such transfers facilitate or delay labor market entry is an open question. On the one hand, unconditional income may allow young adults to search longer rather than accept the first available job, or to invest in training or education. On the other hand, because transfers are not tied to work or training requirements, the income effect may reduce labor supply and delay entry without improving the jobs eventually obtained.

Existing evidence on UBI and UCTs is mixed. In high-income settings, many studies find null or near-zero employment effects, while others document small and context-dependent labor-supply responses (Akee et al., 2010; Hum and Simpson, 1993; Calnitsky et al., 2019; Jones and Marinescu, 2022; Verho et al., 2022; Vivalt et al., 2026). Yet most existing studies rely on short-term experiments, leaving limited evidence on longer-run labor market effects after transfers end,[1] especially for young adults at the point of labor market entry (Hoynes and Rothstein, 2019).

This paper studies the labor market effects of unconditional income support for young adults using the Seongnam Youth Dividend, a universal cash transfer introduced in 2016 by the city of Seongnam, South Korea. The program provided all 24-year-olds who had lived in the city for at least three years with quarterly payments totaling one million KRW over one year.[2] The policy was explicitly motivated by the idea that income support during the transition to adulthood could help young people move into better jobs. Because the program targeted one age cohort in a single city, general-equilibrium effects are unlikely to obscure labor-supply responses. We focus on the first cohorts to receive the full sequence of payments, before the program was extended to other cities in Gyeonggi Province.

To evaluate the program, we combine administrative microdata from Statistics Korea linking employment records and census information from 2015 to 2021. These data allow us to follow the universe of eligible individuals at quarterly frequency and observe both

[1] Only a few studies in the broader cash-assistance literature examine longer-run labor market outcomes after assistance ends; see, for example, Aizer et al. (2020); Deshpande (2016); Price and Song (2026).

[2] This amount is comparable to the Alaska Permanent Fund, one of the first universal basic income programs financed by state oil revenues. Payments in recent years range from $1,000 to $2,000 per year.

individual labor market outcomes and household characteristics. Relative to prior studies of unconditional cash transfers, this setting provides two advantages. First, unlike experimental studies based on selected participants or self-reported outcomes, our data cover the full population of eligible youths and are free from survey nonresponse, consent-based censoring, and self-reporting errors (Vivalt et al., 2026). Second, the panel extends well beyond the transfer period, allowing us to trace employment trajectories for up to three years after the final payment. The closest administrative-data study, the Finnish Basic Income Experiment, follows 2,000 unemployed adults over 2017–2018 (Verho et al., 2022); our setting covers a citywide, youth-targeted program with a longer post-transfer window.

Our empirical strategy compares 24-year-olds in Seongnam with same-age individuals in neighboring cities that were never exposed to the program, using a stacked difference-in-differences and event-study design following Cengiz et al. (2019). We focus on the 1992 and 1993 birth cohorts, the first full birth-year cohorts eligible for all four payments. Because eligibility timing varies by birth quarter, the design aligns individuals by event time relative to the first quarter in which they become eligible. Our main sample consists of youths residing with their parents at the time of eligibility. We impose this restriction because eligibility depended on three years of continuous residence in Seongnam, and our data, which begin in 2015, do not allow us to verify this condition for everyone. Youths living with their parents are less likely to move than those living on their own, so we can assign treatment status with little error. They also account for roughly 80 percent of each cohort, their household income is observed, and they are the group for whom the trade-off between immediate work and longer-run career investment is most relevant.

We find that the Youth Dividend temporarily reduces labor supply during the transfer period. The probability of working declines by about 1 percentage point from a 44 percent baseline, and the decline is concentrated almost entirely in regular employment.[3] Quarterly earnings fall by an average of about 44,720 KRW (about $39 at the 2016 exchange rate), or 17.9 percent of the quarterly transfer. Larger quarterly transfers generate larger reductions, and the implied employment elasticity of about -0.17 is similar in magnitude to that in prior U.S. evidence.[4] After payments end, employment returns toward baseline, and we find no significant difference in employment up to three years after the final payment.

The reduction is concentrated among individuals with no positive labor income during the year preceding eligibility. Within this group, treated individuals have an approximately 7 percent lower employment entry hazard than controls, and their shortfall in cumulative entry narrows after payments end but has not closed by the end of the observation window.

[3]Regular employment refers to wage employment under a contract lasting longer than one month, as distinct from daily and temporary employment with contracts shorter than one month; see Section 3.

[4]The estimates in Vivalt et al. (2026) imply an elasticity of about -0.20; see Section 5.2.

Among those who do enter, treated entrants are 1.4 percentage points more likely to enter mid-sized enterprises, somewhat less likely to enter microenterprises, and have modestly higher entry earnings. These differences are suggestive of a slightly longer search for better jobs, but they are small and we do not place much weight on them. Also, because the comparison is conditional on entry, they may reflect selection into who enters and therefore need to be interpreted with caution. Overall, the transfer produced a temporary reduction in labor supply, driven mainly by previously non-employed youths delaying their entry into employment. Employment in the full sample is unchanged three years after the final payment, although the previously non-employed had not fully caught up by the end of the observation window.

This paper contributes to the literature on unconditional cash transfers by providing population-wide, medium-run evidence for young adults at the point of labor market entry. Existing studies fall into three groups (Appendix Table A.1). The earliest evidence comes from the Mincome negative-income-tax experiment in Dauphin, Canada, in the 1970s and from permanent dividend programs in the United States, the Alaska Permanent Fund Dividend and the Eastern Cherokee casino dividend; these studies found responses ranging from sizable reductions in household labor-force participation to no detectable change in employment (Hum and Simpson, 1993; Calnitsky et al., 2019; Jones and Marinescu, 2022; Akee et al., 2010). Recent randomized controlled trials in Finland, Germany, Spain, and the United States find employment effects that are null or modestly negative for unconditional transfers and larger for the means-tested program in Barcelona (Verho et al., 2022; Bernhard et al., 2025; Verlaat et al., 2025; Vivalt et al., 2026; West and Castro, 2023; Balakrishnan et al., 2024). In developing countries, unconditional transfers sometimes raise participation and earnings through liquidity and entrepreneurship channels (Crosta et al., 2024), a mechanism less relevant for young adults in a high-income labor market. None of these studies follows young adults from the point of labor market entry to well after the transfer period, so they cannot tell whether unconditional income delays or improves entry into work.

Our setting fills this gap in three ways. The transfer went to an entire age cohort at the point of labor market entry rather than to low-income or unemployed adults, so the responses we estimate are those of young people deciding when and where to start working. We observe the full eligible population in administrative records rather than a sample of consenting participants with self-reported outcomes. And we follow recipients for three years after the last payment, whereas most trials report outcomes during or shortly after the transfer period; the few studies of cash assistance that look beyond program end concern other populations (Aizer et al., 2020; Deshpande, 2016; Price and Song, 2026). Together, these features allow us to separate a temporary labor-supply response from a lasting change

in employment trajectories, which short-horizon studies of low-income adults cannot do. Relative to prior Korean studies of the Seongnam and Gyeonggi youth programs, which rely on survey data and short-term outcomes (Yun and Oh, 2021; Yoo et al., 2022), this paper provides, to our knowledge, the first population-wide causal evidence on how unconditional income support shapes early-career labor dynamics.

The rest of the paper proceeds as follows. Section 2 describes the program, Section 3 the data, Section 4 the empirical strategy and sample, Section 5 the results, and Section 6 concludes.

2 Policy Background

In 2016, Seongnam-si, a suburban city in South Korea, introduced an unconditional cash transfer program targeting young adults, known as the Youth Dividend. The program provided quarterly transfers of 250,000 KRW—amounting to 1 million KRW per year[5]—to all 24-year-old residents who had continuously resided in the city for at least three years. The stated objectives were to provide financial support during the transition to adulthood, facilitate labor market entry, and stimulate local economic activity.

Proposed in mid-2015 and formally announced by Seongnam's mayor that September, the policy was scheduled to launch in 2016 pending approval from the Ministry of Health and Welfare. In December 2015, however, the Ministry publicly rejected the proposal, citing legal and policy concerns. This unexpected rejection created substantial uncertainty about the program's implementation, and whether eligible residents would receive the benefit remained unclear until the launch date. Despite the central government's opposition, Seongnam proceeded with implementation in January 2016 and subsequently filed a legal challenge against the Ministry.

As a result of administrative disruptions during the initial implementation, individuals who applied for the Youth Dividend before December 26, 2016 received only 125,000 KRW in their first eligible quarter, with the remaining 125,000 KRW disbursed later as a lump-sum adjustment. Because individuals born in certain quarters of 1992 became eligible before this date, they received half-payments for one or two quarters depending on birth quarter and application timing.[6] The correction ensured that total annual transfers still reached

[5] At the 2016 average exchange rate of approximately 1,160 KRW per USD, 250,000 KRW corresponds to about $215, and 1 million KRW corresponds to about $860.

[6] Applicants born in 1992 had between one and two eligible quarters before December 26, 2016. Those who applied before that date received 125,000 KRW in those early quarters and the withheld balance afterward. The number of affected quarters varied by birth quarter (two for Q1 births and one for Q2 births), whereas Q3 births received a half-payment only if they applied before December 26. Q4 births became eligible in the first quarter of 2017 and therefore received the full payment in every eligible quarter.

1 million KRW (Figure A.3). From the fourth-quarter 1992 birth cohort onward, the full quarterly payment of 250,000 KRW was disbursed without interruption. Because the amount distributed to each birth cohort during eligible quarters varied slightly, the average transfer received differed across event quarters, rising over the first several quarters before declining slightly in the final payment quarter, as shown in Figure A.2. Transfers were issued through prepaid electronic cards that could be used only at small local businesses or for utility payments, excluding large retail chains. This design reflected the program's dual objectives of supporting young adults and stimulating local economic activity. In 2016, during the partial rollout phase, total disbursements amounted to 10.3 billion KRW. Take-up rates were high, reaching 93.9% in 2016 and 96.0% in 2017 according to a government report (Kim et al., 2019). Because eligibility was determined by the calendar quarter in which individuals turned 24, the timing of initial receipt varied by birth quarter, introducing additional variation in exposure across cohorts.

3 Data

This study leverages a newly released set of administrative microdata from Statistics Korea that, for the first time, allow researchers to link the Employment Registry with the Population Census at the individual level. This linkage enables us to track the universe of formal-sector workers and residents over an extended time horizon and observe rich employment-related dynamics—including job transitions, earnings, and migration—rarely available outside the Scandinavian countries (Norway, Finland, Sweden, and Denmark).[7]

The Employment Registry contains universe-level employment records for 2015–2021, compiled from mandatory filings with South Korea's four major social insurance programs and the National Tax Service. The data provide detailed job histories for all formal-sector workers, including unique individual identifiers, date of birth, hire and separation dates, employment status, and annual earnings for each job spell. Information on educational attainment, hours worked, and occupation is not available. Contract types are classified into five categories: (i) daily and temporary employment (contracts shorter than one month), (ii) regular employment (contracts longer than one month), (iii) self-employment, (iv) freelance work, and (v) special contracts, which include platform-based or gig work.

Exact employment start and end dates are observed for daily and temporary employment, regular employment, and self-employment, whereas freelance work and special-contract jobs

[7] To our knowledge, South Korea is the only non-Scandinavian country where individual-level linkage between population and employment registries is available at a national scale. Our team is one of the first to utilize this new dataset since its release by Statistics Korea.

are recorded only on an annual basis. For these categories, individuals are flagged as employed if they held such a job at any point during the year, without information on start or end dates. In addition, annual earnings are unavailable for freelance and special-contract jobs; therefore, we exclude these categories when constructing our employment measures. We use this dataset to construct labor market outcomes such as employment, earnings, and job transitions.

The Population Census is a registry-based database of all residents, observed annually each November, and includes detailed information on residential registration and family relationships. We use these data to determine eligibility for the Youth Dividend based on continuous residency within Seongnam for at least three years prior to benefit receipt, and to identify whether individuals live with their parents and whether they reside in group households such as dormitories and orphanages.

4 Empirical Methodology

We estimate the effects of the Seongnam Youth Dividend on labor market outcomes using a stacked difference-in-differences (DiD) design following Cengiz et al. (2019). Treatment is defined at the birth-quarter–by–city level: individuals become eligible for the transfer in the quarter they turn 24. Because eligibility timing varies across birth quarters, the mapping between calendar time and event time differs across cohorts.

To address this staggered timing, we realign individuals by event time, measured in quarters relative to the first quarter of eligibility, and estimate dynamic treatment effects using fully interacted event-time indicators. The stacked design mitigates biases that can arise in conventional two-way fixed-effects estimators, which may generate negative or non-convex weighting across cohorts and distort treatment effect estimates (Roth et al., 2023). Specifically, each birth-quarter–city pair is treated as a separate event and the corresponding samples are stacked. Treated cohorts are compared with individuals born in the same birth quarter in cities that never implemented the program. The identifying variation therefore comes from comparing individuals who reach age 24 in *Seongnam-si*, where the transfer is implemented, with individuals in neighboring cities where the program was never introduced.

We estimate the following specification:

$$Y_{itj} = \sum_{k=-4}^{15} \beta_k \left(\mathbf{1}\{\text{EventTime}_{it} = k\} \times \text{Treated}_i \right) + \alpha_i + \tau_{tb(i)} + \rho_{jq(t)} + \varepsilon_{itj} \tag{1}$$

Here Y_{itj} denotes the outcome of interest for individual i observed in calendar time t, with $q(t) \in 1, 2, 3, 4$ denoting the corresponding quarter of the year, residing in city j, and born in birth-year-quarter $b(i)$. Event time k is defined relative to the first quarter in which individual i turns 24—the quarter when the individual becomes eligible for the transfer if residing in *Seongnam-si*. Treated_i is an indicator equal to one for individuals residing in *Seongnam-si* prior to turning age 24.

We estimate dynamic treatment effects from four quarters before eligibility ($k = -4$) through fifteen quarters after eligibility ($k = 15$). Following Miller (2023), we normalize the event-time coefficients relative to the average over the four pre-treatment quarters ($k = -4, \ldots, -1$), rather than a single reference quarter. This normalization reduces reliance on fluctuations in any single pre-treatment quarter. Thus, each coefficient β_k represents the estimated treatment effect at event time k relative to the average pre-treatment difference between the treated and control groups. The analysis is conducted using a balanced panel over the full event window $[-4, 15]$, restricting the sample to individuals who are continuously observed throughout this period.

The specification includes three sets of fixed effects: individual fixed effects (α_i), which account for time-invariant individual characteristics; birth-year-quarter–by–calendar-time fixed effects ($\tau_{tb(i)}$), which control for cohort-specific time trends; and city-by-quarter-of-the-year fixed effects ($\rho_{jq(t)}$), which absorb local business-cycle seasonality. Calendar time is measured in quarters, and standard errors are clustered at the individual level.

Identification relies on the parallel trends assumption that, absent the program, outcomes for treated and control individuals would have evolved similarly. We assess this assumption by examining whether the normalized pre-treatment coefficients exhibit systematic deviations from zero.

To summarize treatment effects over broader time horizons, we also estimate a difference-in-differences specification that aggregates event-time indicators into annual periods relative to the first receipt of the transfer:

$$\begin{aligned} Y_{itj} = \beta_0 \left(\mathbf{1}\{\text{During}_t\} \times \text{Treated}_i\right) + \sum_{s=1}^{3} \beta_s \left(\mathbf{1}\{\text{After } s \text{ yr}_t\} \times \text{Treated}_i\right) \\ + \alpha_i + \tau_{tb(i)} + \rho_{jq(t)} + \varepsilon_{itj} \end{aligned} \tag{2}$$

$\mathbf{1}\{\text{During}_t\}$ indicates the four quarters during which individuals are age 24 and receive the transfer. $\mathbf{1}\{\text{After } s \text{ yr}_t\}$ denotes indicators for the s-th year after the final payment, where $s = 1, 2, 3$ correspond to the four quarters in the first, second, and third years following the final payment. The omitted category is the four quarters preceding the start of transfers. The

coefficients capture the average effects during the transfer year and in each of the subsequent three years relative to the pre-transfer period.

4.1 Main Analysis Sample

Our main analysis sample consists of individuals born in 1992 and 1993, the first full birth-year cohorts for which all birth quarters were eligible to receive all four quarterly payments under the Seongnam Youth Dividend program.[8] The unit of observation is an individual-by-quarter. We observe individuals from four quarters before the quarter they turn 24 through fifteen quarters afterward.

The treatment group consists of residents of *Seongnam-si*. The control group includes residents of neighboring cities within *Gyeonggi Province* that border *Seongnam-si*: *Hanam-si, Yongin-si, Gwacheon-si, Suwon-si, Uiwang-si*, and *Gwangju-si*.[9]

Eligibility required three years of continuous residency in the municipality. Because our administrative data begin in 2015, we cannot fully verify this condition for cohorts receiving payments beginning in 2016. We therefore impose the residency condition as much as possible by requiring individuals to reside continuously in the same city for one or two years prior to eligibility depending on the cohort. The same restriction is applied to individuals in control cities.

We focus the main analysis on individuals residing with their parents at the time of first receipt of the transfer, identified as children of the household head in the Population Census. This group represents roughly 80% of each birth cohort. Focusing on co-residing individuals is motivated by concerns related to the program's residency requirement. Individuals living with their parents are likely to have more stable residential arrangements than those living independently and are therefore more likely to satisfy the three-year residency requirement, reducing potential measurement error in treatment assignment based on residency. This group also represents a large and policy-relevant segment of young adults at the age of eligibility.

[8] Individuals born in 1991 were eligible for the program to varying degrees depending on their birth quarter because the program began in January 2016 and eligibility was restricted to residents aged 24. Although all 1991 birth-quarter cohorts received at least one payment, only those born in the fourth quarter remained eligible for all four quarterly payments, while earlier birth-quarter cohorts aged out before completing the full payment sequence. We therefore focus on the 1992 and 1993 cohorts, who were the first to receive the full sequence of payments. We exclude cohorts born in 1994 or later because the program was later extended to other cities in Gyeonggi Province, making individuals in the control cities eligible for transfers during the observation period.

[9] We use neighboring cities rather than the entire *Gyeonggi Province* as the primary control group to improve comparability. As shown in Appendix Table A.3, average household earnings in the broader province are substantially lower than those in both *Seongnam-si* and its bordering cities, making the province a less suitable comparison group.

Finally, we construct a balanced panel of individuals observed continuously from four quarters before to fifteen quarters after eligibility. Applying these restrictions yields 15,851 treated individuals and 51,254 control individuals in the main sample.

4.2 Outcome Variables

Our primary outcomes capture employment and earnings. We report three measures: (1) *any labor income*, (2) *regular employment*, and (3) *quarterly earnings from regular jobs*.

We define *any labor income* as an indicator equal to one if an individual has positive earnings in a given quarter. This measure encompasses self-employment and wage employment in either daily or temporary jobs or regular jobs. Throughout the paper, we refer to this measure as employment, capturing work that generates observed positive labor income. As described in Section 3, freelance and special-contract jobs are excluded from our employment measures because exact employment start and end dates are unavailable and annual earnings are not reported for these categories. We define *regular employment* as employment under a contract lasting longer than one month, consistent with the definition provided in Section 3. Quarterly earnings from regular jobs are calculated using reported job start and end dates together with annual earnings. If an individual holds multiple regular jobs within a quarter, earnings are aggregated across all such jobs. To limit the influence of extreme values, quarterly earnings are top-coded at the 99th percentile of the pooled sample.

4.3 Descriptive Statistics

Table 1 reports observable characteristics of treated and control individuals measured in the quarter prior to program eligibility. Individuals in *Seongnam* and those in the control cities appear broadly similar in their demographic and employment characteristics. Gender composition is nearly identical, with women accounting for approximately half of each group. About 44% of treated individuals and 46% of control individuals worked in the quarter before turning age 24. Most employment took the form of regular employment, defined as jobs with contracts lasting longer than one month: 38% of treated individuals and 40% of control individuals held regular jobs.

Youth in *Seongnam* earned on average about 139,600 KRW less per quarter than those in the control cities (approximately $123 in 2015 USD).[10] Per capita household income levels, however, are similar across groups: average annual household income is 20 million KRW among treated individuals and 18.8 million KRW among controls (a difference of approximately $1,116 in 2015 USD), and the difference is not statistically significant.

[10] 139,600 KRW corresponds to approximately $123 using the 2015 average exchange rate.

Finally, mobility rates are similar across groups. About 6% of treated individuals and 7% of control individuals move to another city in the year they turn 24, and the difference is not statistically significant. These comparable migration rates provide little evidence of differential sorting across treated and control cities prior to program eligibility.

5 Impacts on Youth Labor Market Outcomes

5.1 Effects on Labor Supply During and After the Transfer Period

We begin by examining the effects of the Youth Dividend on labor supply. Employment declines during the quarters in which individuals receive the Youth Dividend and rebounds once payments end (Figure 1).[11] Event-study estimates indicate that statistically significant reductions in labor supply are concentrated during the transfer period. After transfers end, the point estimates point to a gradual recovery toward baseline, suggesting that the labor-supply response is temporary rather than persistent.

Averaged across the four quarters of transfer receipt, the probability of working declines by 1 percentage point from a baseline of 44%, corresponding to a 2.3% reduction in employment (Table 2, Column 1). The decline in overall employment appears to be concentrated primarily in regular employment. The probability of regular employment declines by 0.9 percentage points from a baseline of 38%, corresponding to a 2.4% reduction (Table 2, Column 2).[12]

Quarterly earnings also decline modestly during the transfer period. The average decline in quarterly earnings is 44,720 KRW relative to a baseline of 1.90 million KRW, corresponding to a 2.3% reduction (Table 2, Column 3). Given a quarterly transfer of 250,000 KRW, the estimated reduction in quarterly earnings amounts to approximately 17.9% of the transfer.

These effects appear short-lived. After the transfer period, employment probabilities gradually move back toward their pre-program levels, and difference-in-differences estimates show no statistically significant changes in employment up to three years after the final payment (Table 2). Quarterly earnings show a broadly similar recovery pattern, although the third-year estimate is modestly positive and statistically significant at the 10 percent level.[13] At the quarterly level, the event-study point estimates for earnings also become positive at longer

[11] The results are similar when we expand the sample to include individuals who do not reside with their parents (Appendix Figure A.5).

[12] Appendix Figure A.6 reports separate event-study estimates for temporary and daily employment and self-employment. These outcomes show little evidence of meaningful changes during the transfer period.

[13] In the baseline specification, quarterly earnings are top-coded at the 99th percentile. When observations above the 99th percentile are instead set to missing, the third-year difference-in-differences estimate is no longer statistically significant.

horizons (Figure 1c), but they are imprecisely estimated. We therefore interpret these later increases as suggestive rather than causal evidence of longer-run improvements in job quality, and find little evidence that the temporary reduction in earnings persists after payments end.

These results are not driven by any single control city. Excluding each control city in turn yields estimates that are broadly similar to the baseline results (see Appendix Figure A.7). In addition, when we expand the control group to include adjacent districts within Seoul, the overall patterns remain broadly similar. Specifically, declines during the transfer period are followed by a partial recovery over the first two years after payments end, although the estimates differ somewhat at longer horizons (see Appendix Figure A.8).

5.2 Continuous Treatment Effects and Labor-Supply Elasticity

To examine how the labor-supply response scales with the size of the transfer, we exploit variation in the amount of cash received across cohorts. As described in Section 2, the 1992 birth cohort experienced uneven quarterly payments due to administrative delays: Early cohorts received half payments for one or two initial quarters and a lump-sum payment in the subsequent quarter (Figure A.3). In contrast, the payment schedule had normalized by the fourth-quarter 1992 cohort, and all 1993 birth-quarter cohorts received a uniform 250,000 KRW per quarter.

To leverage this variation, we replace the binary treatment indicator with the actual transfer amount received in a given quarter, measured in units of 1 million KRW, and estimate the same specification as in Equation 2. The transfer variable is set to zero for control individuals and for treated individuals outside the payment period. The coefficient therefore measures the change in outcomes associated with receiving an additional 1 million KRW in transfers in a given quarter.

These estimates closely track the binary-treatment results in Table 2. An additional 1 million KRW in transfers reduces the probability of working by 3.3 percentage points, with the reduction appearing to operate primarily through regular employment, which declines by 3.8 percentage points (Table 2, Panel B). Quarterly earnings also decline significantly, by approximately 255,400 KRW for each additional 1 million KRW in transfers. Together, these estimates suggest that the short-run reductions in employment and earnings scale with the amount of the transfer.

To assess the economic magnitude of the main employment effect and facilitate comparison with previous studies, we translate the binary-treatment estimate in Panel A into a

short-run labor-supply elasticity. The implied elasticity is approximately -0.17 for any labor income and -0.18 for regular employment.[14]

This estimate is broadly consistent with prior evidence. Vivalt et al. (2026) find that an unconditional transfer of $12,000 over three years—about 56% of annual baseline income—reduced employment by 4.2 percentage points from a 63% baseline, implying an elasticity of -0.20 when measured relative to net-of-tax income. Although the programs differ substantially in transfer size and target population, the implied elasticities are similar in magnitude. This finding suggests that young adults who are still transitioning into stable employment exhibit a meaningful short-run labor-supply response even to a relatively modest unconditional transfer.

5.3 Heterogeneous Effects by Prior Employment Status

The temporary reduction in labor supply documented above could reflect either delayed labor market entry among previously non-employed individuals or temporary labor-force withdrawal among those already working. To distinguish between these possibilities, we examine heterogeneity by prior employment status.

Figure 2 plots event-study estimates from Equation 1 separately for individuals with and without prior employment, and Appendix Table A.4 reports the corresponding estimates aggregated into the transfer period and post-transfer years. Prior employment is defined as having positive labor income in at least one quarter during the year preceding eligibility. Each point represents the change relative to the corresponding subgroup among control individuals.

The decline in labor supply is concentrated primarily among individuals without prior employment. During the transfer period, the probability of working declines by 2.3 percentage points among individuals who had no positive labor income during the preceding year (Appendix Table A.4, Column 1). The corresponding decline in regular employment is 1.7 percentage points (Column 2), while quarterly earnings decrease by approximately 90,530 KRW (Column 3). These effects are also visible in Figure 2, which shows a clear decline in employment and earnings for the previously non-employed group beginning with transfer receipt.

In contrast, the estimated employment effects are substantially smaller among individuals with prior employment. During the transfer period, overall and regular employment each

[14] We calculate the labor-supply elasticity as $\eta_e = \frac{\bar{Y}}{\partial Y}\frac{\partial p}{\bar{p}}$, where $\bar{Y}$ is baseline individual quarterly earnings (1.90 million KRW), ∂Y is the increase in income from the quarterly transfer (250,000 KRW), ∂p is the estimated change in the employment probability, and $\bar{p}$ is the corresponding baseline employment rate. Calculations use the rounded estimates reported in the text.

decline by approximately 0.5 percentage points in this group, but neither estimate is statistically significant. quarterly earnings also decline, by approximately 10,390 KRW, although this estimate is not statistically significant. These results provide little evidence that the transfer induced temporary labor-force withdrawal among individuals with prior employment experience. However, the earnings effects are less clearly differentiated across the two groups than the employment effects.

Taken together, these findings suggest that the reduction in labor supply was driven primarily by delayed entry into employment among individuals without recent employment experience, rather than by temporary labor-force withdrawal among those who were already working. At the same time, the combination of relatively stable employment and lower earnings among previously employed individuals during the transfer period provides suggestive evidence of adjustment along the intensive margin, such as reduced hours worked while remaining employed. However, the earnings estimates are not statistically significant, and hours worked are not observed in the administrative data, limiting our ability to assess this mechanism. The persistence of the employment gap after payments end is consistent with a delayed transition into employment, although the subsequent recovery in earnings makes it difficult to interpret this pattern as evidence of a lasting deterioration in labor market outcomes.

5.4 Effects on the Timing and Characteristics of First Jobs

To investigate whether the employment decline among individuals without prior employment reflects delayed entry, we examine the timing of their first employment after eligibility. We then assess whether the Youth Dividend is associated with differences in the characteristics of these jobs. Accordingly, we restrict the sample to individuals with no positive labor income during the year preceding eligibility.

First, we use Cox proportional hazards models to examine the time from eligibility to the first observed entry into employment. Individuals who do not enter by the end of the observation period are right-censored. All models control for average household income during the baseline period and a female indicator, and include birth-year-quarter fixed effects. Table 3 reports the results for entry into any job with positive labor income and into wage employment. For entry into employment, defined as having positive labor income, the estimated treatment coefficient is -0.072, implying a hazard ratio of 0.93. Among individuals who have not yet entered employment, the entry hazard is approximately 7 percent lower in the treatment group than in the control group. Similarly, the coefficient for entry into wage employment is -0.064, corresponding to a hazard ratio of 0.94 and an approximately 6

percent lower entry hazard.[15] These results are consistent with the delayed-entry interpretation discussed in Section 5.3. Similarly, we examine how the entry gap evolves over time by applying the main event-study specification in Equation 1 to cumulative entry into employment. Appendix Figure A.9 shows a pattern consistent with delayed entry. Following the first payment, a negative gap in cumulative entry emerges and widens during the transfer period. The gap persists after payments end and narrows at longer horizons, but remains below zero through the end of the observation period. Together with the lower entry hazard estimated in the Cox models, this pattern suggests that delayed entry into employment contributes to the reduction in labor supply documented in the main analysis.

Second, we ask whether the delayed entry documented above is accompanied by differences in the jobs obtained, comparing employer size and quarterly earnings in the first quarter with positive labor income after the first payment.[16] The sample is restricted to individuals who enter employment by the end of the observation period. As in the Cox models, we regress each first-job characteristic on treatment status, controlling for a female indicator, baseline household income, and birth-year-quarter fixed effects. We add calendar-quarter fixed effects for the quarter of entry, which absorb the two time dimensions of our main specification. Standard errors are clustered at the city $\times$ birth-year-quarter level, the level at which treatment is assigned. Table 4 reports the results. Treated entrants are 1.4 percentage points more likely than control entrants to enter mid-sized enterprises, while the probability of entering a large firm does not differ. They are also 1.0 percentage point less likely to enter microenterprises and have entry earnings higher by approximately 65,220 KRW, an economically small difference; both estimates are significant only at the 10 percent level. These differences are suggestive of a slightly longer search for jobs at somewhat larger firms with higher earnings, but they are small and we do not place much weight on them. Also, because the comparison is conditional on entering employment, they may reflect selection into who enters rather than a change in the jobs available to a given individual, and the two explanations cannot be separated in this design.

[15] The proportional hazards assumption is not rejected at conventional levels for entry into any labor income with robust standard errors, is rejected only marginally with standard errors clustered at the city $\times$ birth-year-quarter level, and is clearly rejected for wage employment with clustered standard errors (Table 3). A rejection indicates that the hazard ratio varies with time since eligibility, so the reported coefficients should be read as averages over the observation window. Such variation is expected if the gap in entry is concentrated in the transfer period and narrows afterward, as the cumulative-entry estimates in Appendix Figure A.9 suggest.

[16] Firm size categories follow the Business Register classification, based on industry-specific sales thresholds, assets, and ownership structure, with additional employment thresholds used to identify microenterprises.

6 Conclusion

This paper studies the labor market effects of unconditional income support for young adults using the Seongnam Youth Dividend program. By following eligible individuals for up to three years after the final payment, the study provides rare evidence on the medium-run labor market effects of youth-targeted unconditional cash transfers. Using population-wide administrative data and a stacked difference-in-differences design, we find that the transfer led to a temporary reduction in labor supply during the payment period. The decline was concentrated in regular employment, and post-transfer point estimates indicate a gradual return toward baseline after payments end. The reduction in labor supply is most pronounced among individuals with no positive labor income during the year preceding eligibility. Within this group, treated individuals have a lower employment entry hazard and persistently lower cumulative entry probabilities than controls, consistent with delayed entry into employment.

The results point to a delay in labor market entry rather than a lasting withdrawal. In the full sample, employment returns toward baseline after payments end; among individuals without prior employment, the shortfall in cumulative entry narrows at longer horizons but has not closed by the end of the observation window. Among previously non-employed individuals who enter employment, treated entrants are somewhat more likely to work at mid-sized firms and have modestly higher entry earnings. These differences are suggestive of a slightly longer search for better jobs, but they are small, and because the comparison is conditional on entry they may reflect selection into who enters; we therefore interpret them with caution. The administrative data do not reveal how recipients used the transfers; in a Seongnam City survey, the most common reported uses were groceries and meals, gifts to parents (38%), leisure activities (22.5%), and job-search-related expenses (22.1%) (Kim et al., 2019).

The external validity of these findings is limited in two respects. The policy targeted a single age cohort in one city, and the analysis covers the early years of the program. In addition, because the program covered a small and specific population, the estimates capture individual labor-supply responses rather than the general-equilibrium effects that a larger or longer-running program might generate. Within this setting, however, the estimates do not hinge on the choice of comparison cities: the transfer-period estimates are similar in magnitude when each control city is excluded in turn and when the control group is expanded to adjacent districts in Seoul, although the post-transfer estimates in the expanded sample differ somewhat at longer horizons. The evidence therefore suggests that a modest unconditional transfer can temporarily reduce regular employment among young adults but

may be insufficient on its own to generate sustained improvements in early-career labor market outcomes.

Future research could extend these findings in several directions. First, the administrative data record earnings and employer size but not occupation, hours worked, or other firm characteristics, so whether the transfer helped young adults obtain better job matches remains open. Second, because the transfers were distributed as a local currency redeemable only within Seongnam-si, the program may have affected local consumption, employment, and business activity; these general-equilibrium effects are outside the individual-level estimates reported here and are needed for an assessment of the program's overall welfare effects.

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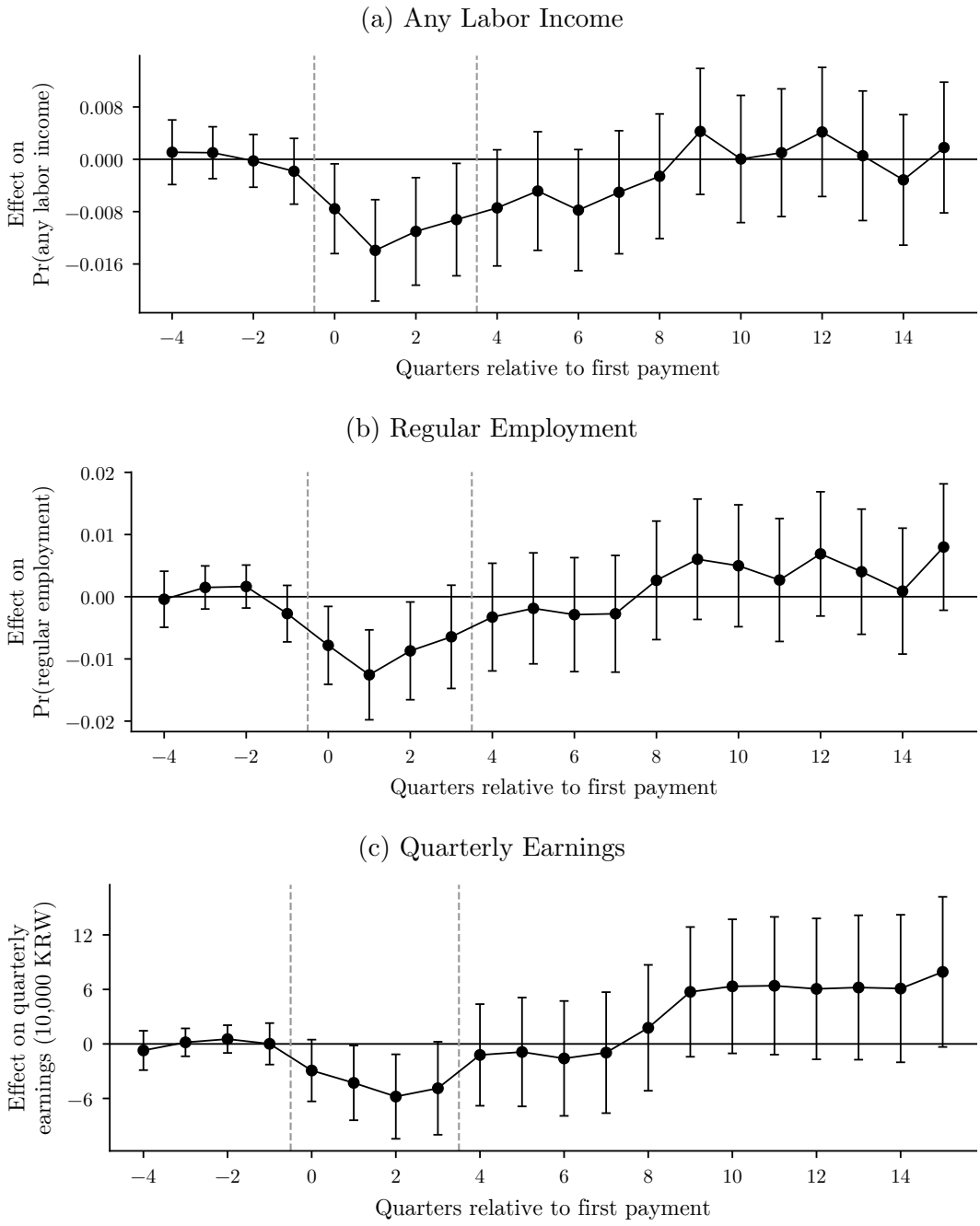
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Figures

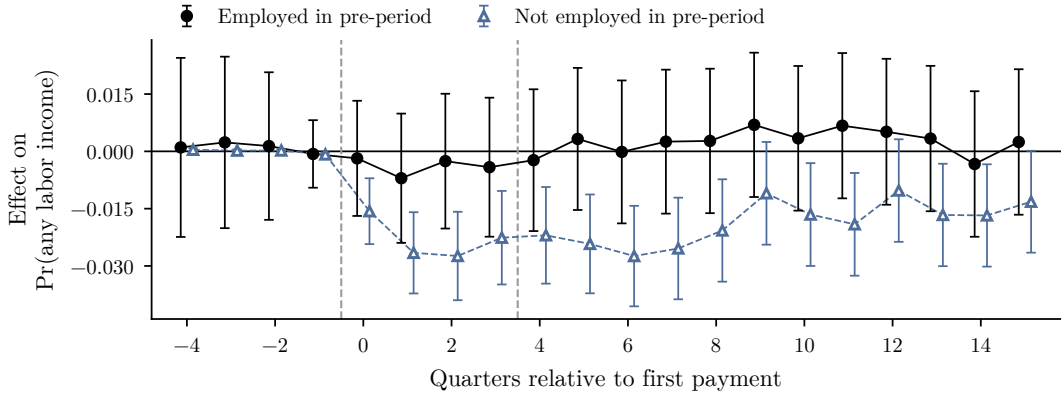
Figure 1: Effects of Cash Transfer on Labor Supply



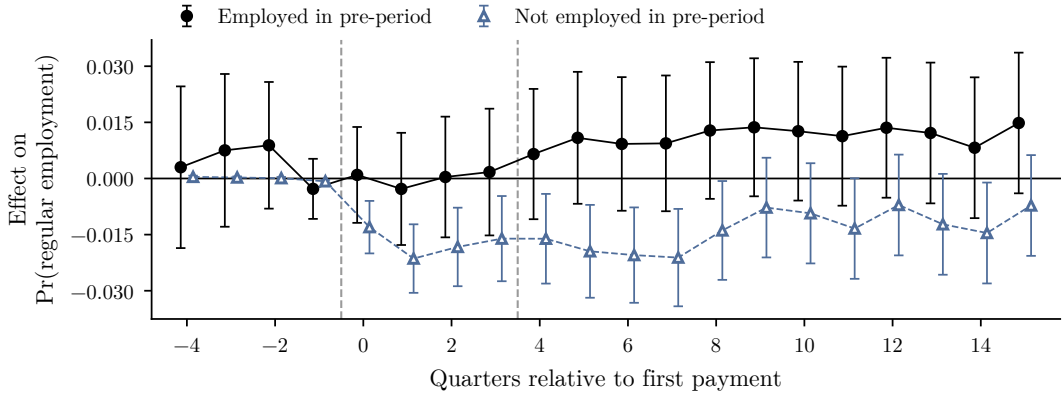
Notes: The figures present dynamic difference-in-differences estimates of the impact of the Youth Dividend on labor market outcomes, including all employment statuses (Panel (a)), regular employment (Panel (b)), and quarterly earnings from regular jobs (Panel (c)). Quarterly earnings in Panel (c) are reported in units of KRW 10,000 and are top-coded at the 99th percentile. The analysis sample includes individuals from the 1992 and 1993 birth cohorts residing with their parents in Seongnam or control cities, restricted to those continuously observed in the panel. The specification includes individual, birth-year-quarter × calendar time, and city × quarter fixed effects. Each dot represents the interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the individual level.

Figure 2: Effects of Cash Transfer on Labor Supply by Previous Employment Status

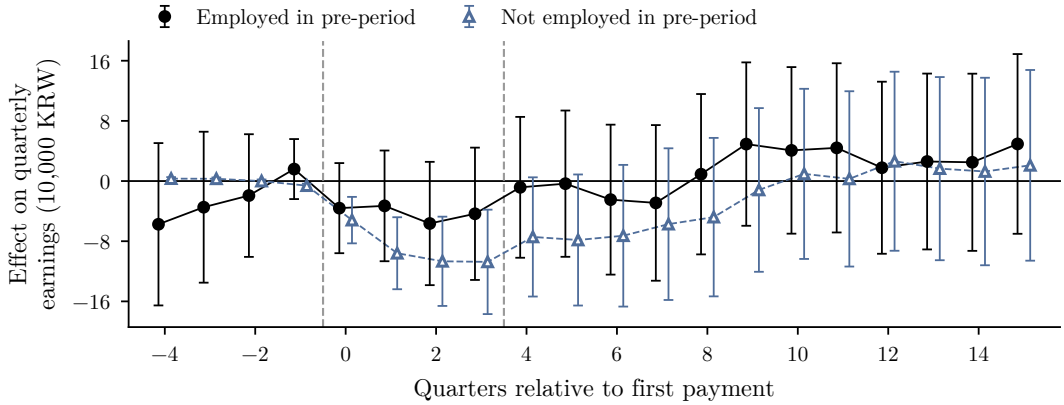
(a) Any Labor Income



(b) Regular Employment



(c) Quarterly Earnings



Notes: The figures present dynamic difference-in-differences estimates of the impact of the Youth Dividend on labor market outcomes, including all employment statuses (Panel (a)), regular employment (Panel (b)), and quarterly earnings from regular jobs (Panel (c)). Quarterly earnings in Panel (c) are reported in units of KRW 10,000 and are top-coded at the 99th percentile. The sample and fixed effects are identical to those in Figure 1, and the analysis further conducts subgroup estimations based on whether individuals were ever employed prior to receiving the benefit. Each dot represents the interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the individual level.

Tables

Table 1: Descriptive Statistics of the Main Analysis Sample

	(1) Seongnam	(2) Neighboring control cities	(3) Difference	(4) Standardized difference
Female	0.508	0.515	-0.008	0.016
	(0.500)	(0.500)	(0.005)	
Any labor income	0.443	0.461	-0.018	0.036
	(0.497)	(0.498)	(0.011)	
Regular employment	0.382	0.402	-0.020	0.041
	(0.486)	(0.490)	(0.011)	
Quarterly earnings (10,000 KRW)	190.33	204.29	-13.96**	0.049
	(278.85)	(288.23)	(4.80)	
Per capita HH income (10,000 KRW)	2,004.11	1,877.92	126.19	-0.039
	(3,857.53)	(2,464.58)	(73.83)	
Migration	0.057	0.065	-0.008	0.033
	(0.231)	(0.246)	(0.006)	
Number of individuals	15,851	51,254		

Notes: Means are reported with standard deviations in parentheses. The sample is the balanced quarterly panel of individuals born in 1992–93 who co-resided with their parents. Quarterly earnings are from regular employment and top-coded at the 99th percentile. Column (1) is the treated city, Seongnam; column (2) the six neighboring Gyeonggi cities used as the main control group. Column (3) reports the difference in means, column (1) minus column (2), with its standard error in parentheses; column (4) the standardized mean difference. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 2: Effects of Cash Transfer on Earnings and Employment Outcomes

	(1) Any labor income	(2) Regular employment	(3) Quarterly earnings (10,000 KRW)
Panel A. Difference-in-differences by period			
Treated × During transfer	-0.010***	-0.009***	-4.47**
	(0.003)	(0.003)	(1.93)
Treated × 1 year after	-0.006	-0.003	-1.16
	(0.004)	(0.004)	(2.89)
Treated × 2 years after	0.001	0.004	5.06
	(0.004)	(0.005)	(3.50)
Treated × 3 years after	0.001	0.005	6.57*
	(0.005)	(0.005)	(3.90)
Panel B. Effect of the transfer amount			
Quarterly transfer amount (million KRW)	-0.033***	-0.038***	-25.54***
	(0.011)	(0.011)	(7.74)
Control mean, year before transfer	0.414	0.366	180.29
Observations	1,342,100	1,342,100	1,342,100
Individual FE	Yes	Yes	Yes
City × quarter FE	Yes	Yes	Yes
Birth cohort × quarter FE	Yes	Yes	Yes

Notes: The sample is the balanced quarterly panel of individuals born in 1992–93 who co-resided with their parents in Seongnam (treated) or the six neighboring control cities; 1,342,100 person-quarter observations. Column (1) is an indicator for any labor income, column (2) an indicator for regular employment, and column (3) quarterly earnings from regular employment in 10,000 KRW, top-coded at the 99th percentile. Panel A reports coefficients on the interaction of a Seongnam indicator with period indicators: the four quarters in which the dividend was paid and each of the three following years; the omitted period is the year before the first payment. Panel B replaces the period indicators with the transfer amount received in the quarter, in million KRW. All specifications include individual, city × calendar quarter, and birth year-quarter × calendar quarter fixed effects. The control mean is the mean of the outcome among control-city individuals in the year before the first payment. Standard errors clustered at the individual level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 3: Entry into Any Labor Income: Cox Proportional Hazards Estimates

	(1)	(2)	(3)	(4)
	Any labor income		Wage employment	
Treated	-0.072***	-0.072***	-0.064***	-0.064***
	(0.015)	(0.009)	(0.015)	(0.009)
Controls	Yes	Yes	Yes	Yes
Birth year-quarter FE	Yes	Yes	Yes	Yes
Standard errors	Robust	City × BYQ	Robust	City × BYQ
PH test (p-value)	0.300	0.090	0.230	0.001
Observations	28,821	28,821	28,821	28,821

Notes: Cox proportional hazards estimates for individuals with no employment during the year before eligibility; 28,821 individuals. The outcome is the quarter of first entry into any labor income (columns (1)–(2)) or into wage employment (columns (3)–(4)); individuals who have not entered by the end of the observation window are right-censored. Log hazard coefficients on the Seongnam indicator are reported. All models control for a female indicator and baseline household income and include birth year-quarter fixed effects. Standard errors in parentheses are heteroskedasticity-robust (columns (1) and (3)) or clustered at the city × birth year-quarter (BYQ) level (columns (2) and (4)). The PH test row reports the p-value of the test of the proportional hazards assumption. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 4: First Job Characteristics

	(1) Quarterly earnings (10,000 KRW)	(2) Large firm	(3) Medium firm	(4) Small firm	(5) Microenterprise
Treated	6.52*	0.004	0.014***	0.002	-0.010*
	(3.61)	(0.003)	(0.004)	(0.008)	(0.006)
Controls	Yes	Yes	Yes	Yes	Yes
Birth year-quarter FE	Yes	Yes	Yes	Yes	Yes
Calendar quarter FE	Yes	Yes	Yes	Yes	Yes
Observations	23,457	23,457	23,457	23,457	23,457

Notes: Characteristics of the first job, measured in the first quarter with positive labor income after the first payment, among individuals with no employment during the year before eligibility who enter employment during the observation window; 23,457 individuals. Column (1) is quarterly earnings in 10,000 KRW, top-coded at the 99th percentile; columns (2)–(5) are indicators for the size class of the employer. All regressions control for a female indicator and baseline household income and include birth year-quarter and calendar quarter fixed effects, the latter defined by the quarter in which the first job begins. Standard errors clustered at the city × birth year-quarter level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

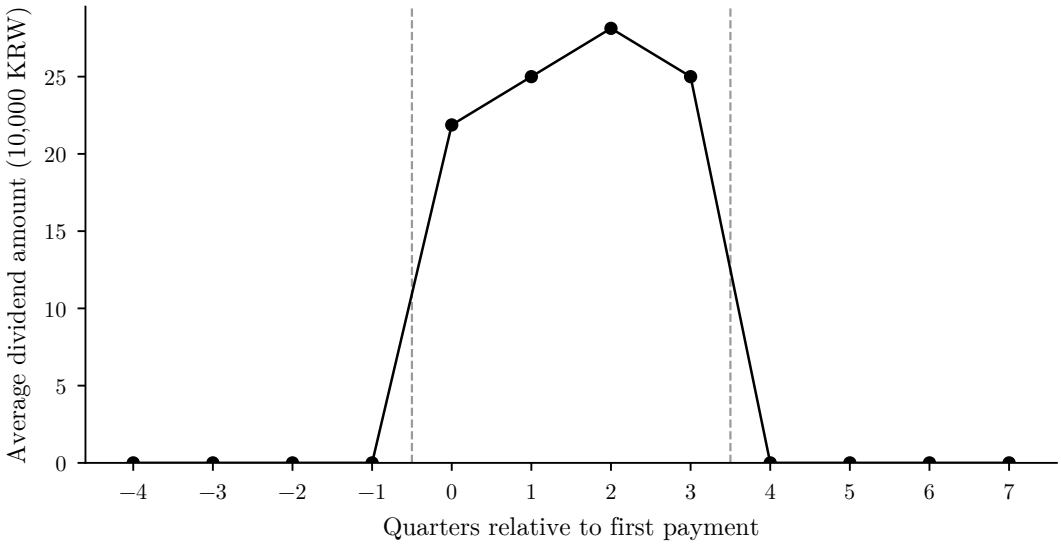
Appendix

Figure A.1: Map of Treated and Control Cities in Gyeonggi Province



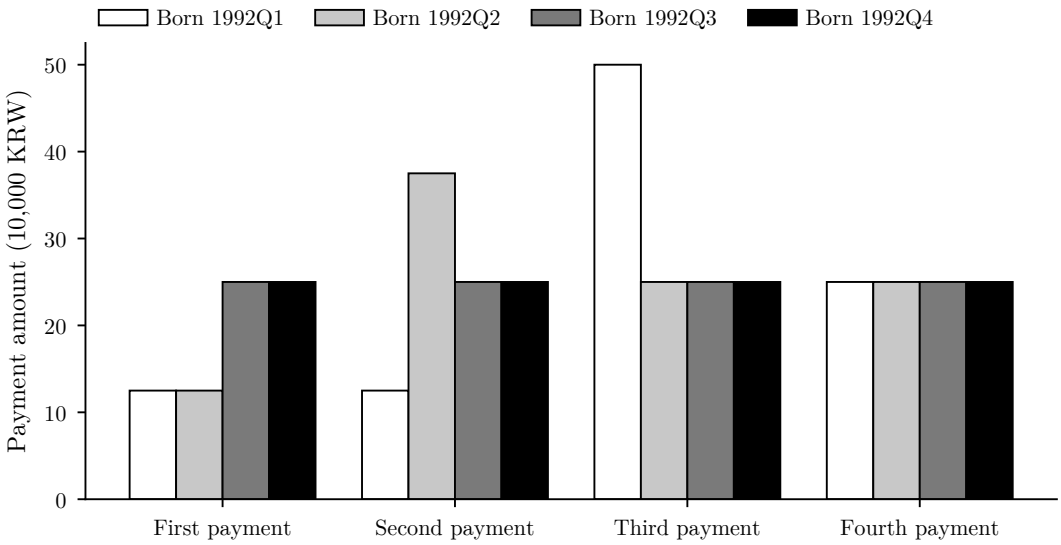
Notes: This figure shows a map of Gyeonggi Province, highlighting the cities used in the main analysis. The darkest-shaded area represents Seongnam-si, the treated city that implemented the Youth Dividend program. The surrounding labeled cities with lighter shading constitute the control group in the main analysis. In addition, the dotted areas indicate an expanded control group that includes neighboring districts within the Seoul Metropolitan Area, which is used for robustness checks.

Figure A.2: Average Dividend Amount



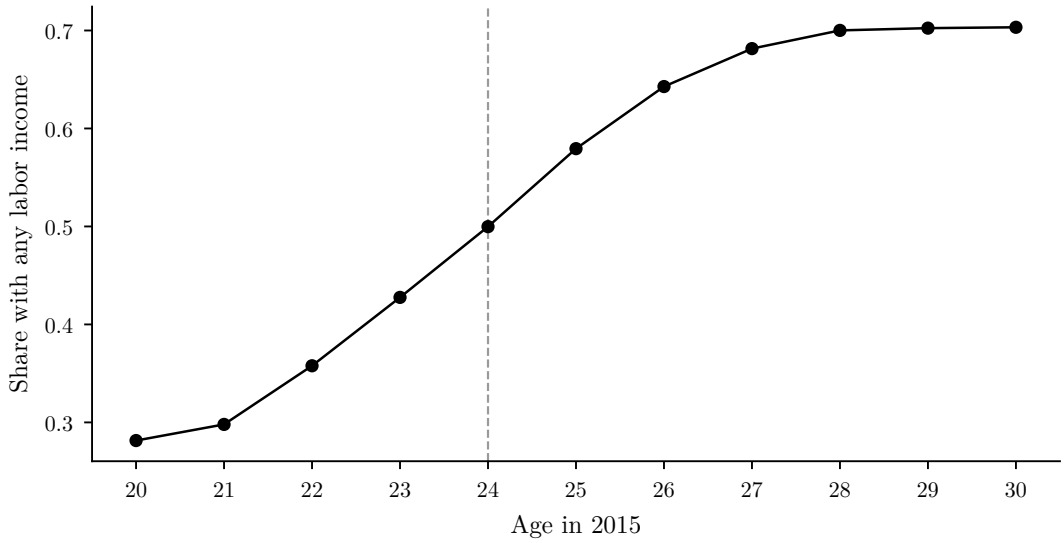
Notes: This figure plots the average dividend amount for the treated group by quarter relative to cash receipt.

Figure A.3: Quarterly Transfer Amounts by Birth Quarter (1992 Cohort)



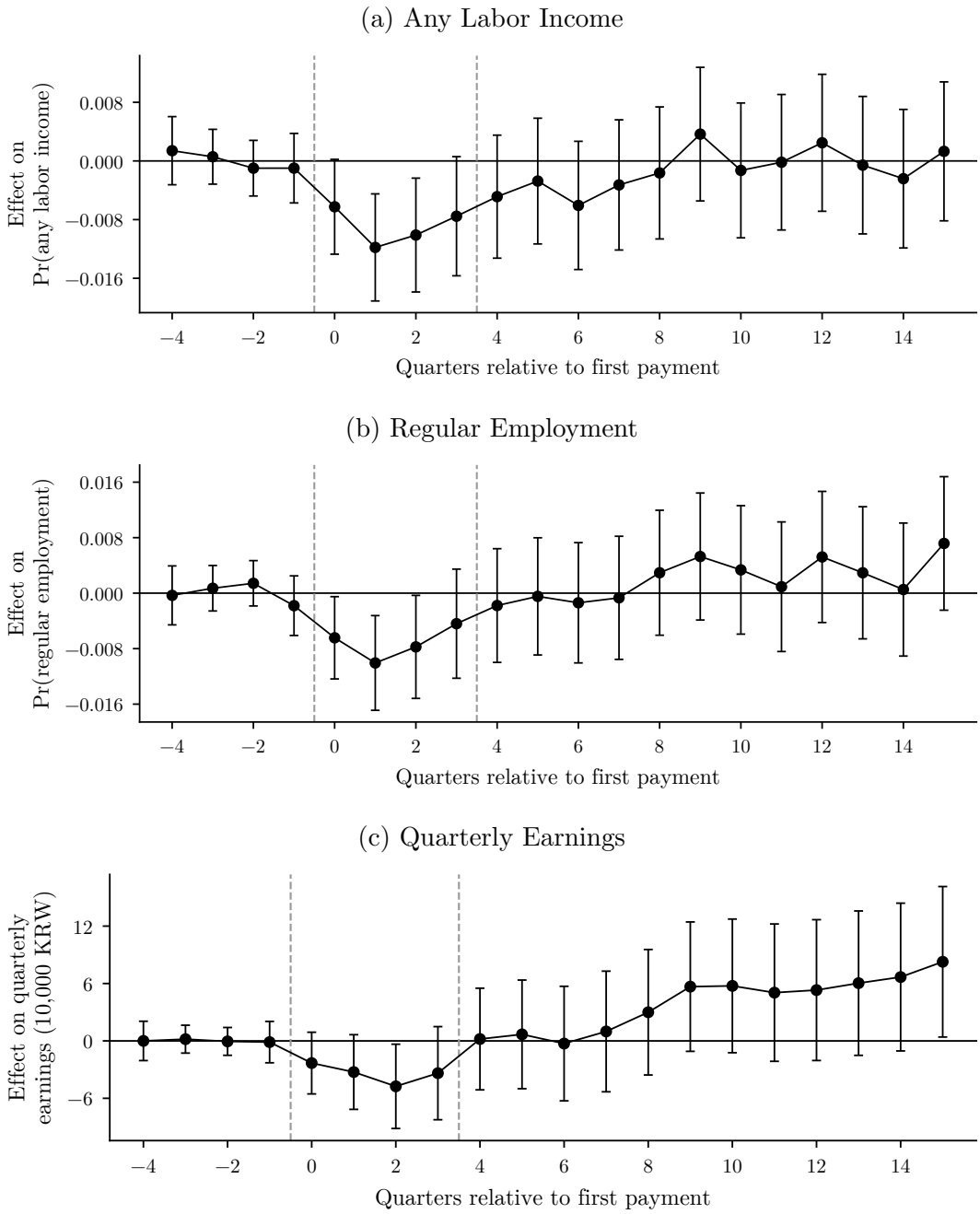
Notes: This table reports the quarterly payment amounts received by individuals born in each quarter of 1992 under the Seongnam Youth Dividend program. Due to administrative delays and legal uncertainty during the program's initial rollout, individuals born in earlier quarters received partial payments in early 2016, followed by lump-sum reimbursements in December 2016 to ensure the total annual transfer reached 1 million KRW. From 2017 onward, quarterly payments of 250,000 KRW were disbursed as originally planned. Individuals in the 1993 cohort received regular quarterly payments of 250,000 KRW without disruption. The information is based on administrative records from Seongnam City.

Figure A.4: Employment by Age (2015)



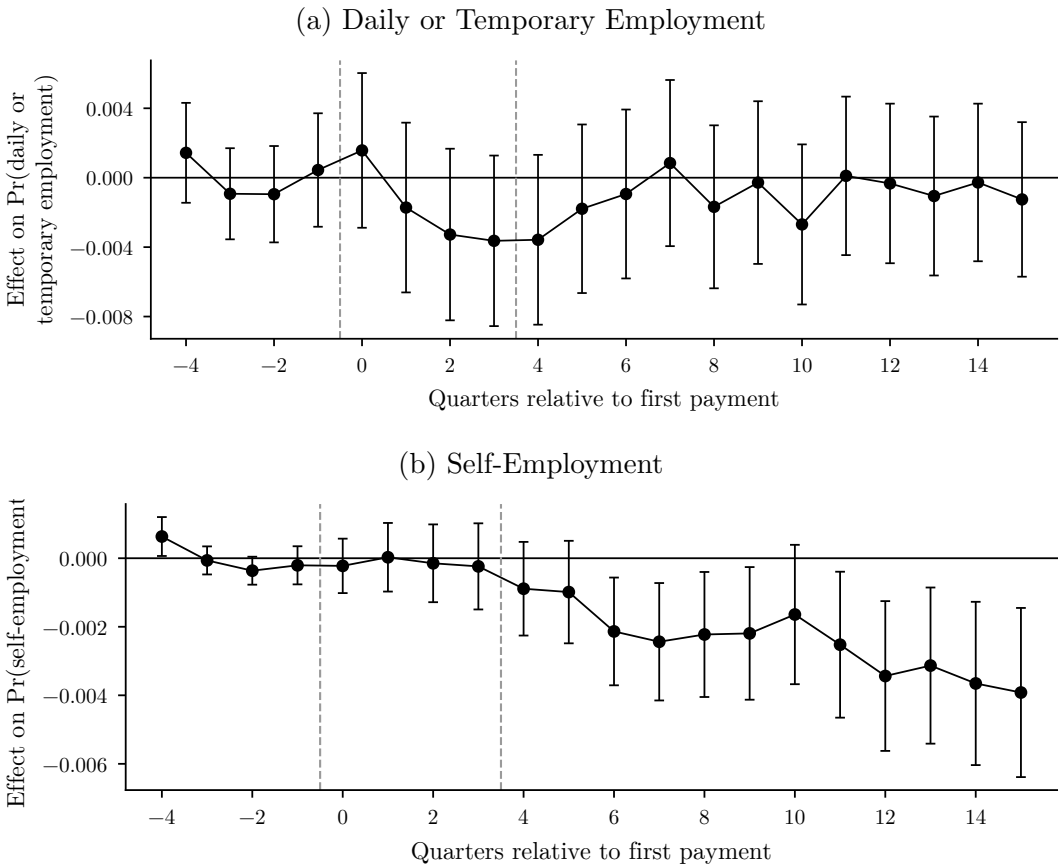
Notes: This figure plots the share of individuals with any labor income by age in 2015, the year prior to the introduction of the Seongnam Youth Dividend program, for both Seongnam and control groups.

Figure A.5: Effects of Cash Transfer on Labor Market Outcomes: Full Sample



Notes: The figures present dynamic difference-in-differences estimates of the effects of the Youth Dividend on employment outcomes, including employment across all employment statuses (Panel (a)), regular employment (Panel (b)), and quarterly earnings from regular jobs (Panel (c)). Quarterly earnings in Panel (c) are reported in units of KRW 10,000 and are top-coded at the 99th percentile. The specification includes the same fixed effects as Figure 1, but the sample is expanded to include both individuals co-residing with their parents and those living separately. Each dot represents the estimated interaction coefficient between treatment status and event time, and the capped lines indicate 95% confidence intervals. Standard errors are clustered at the individual level.

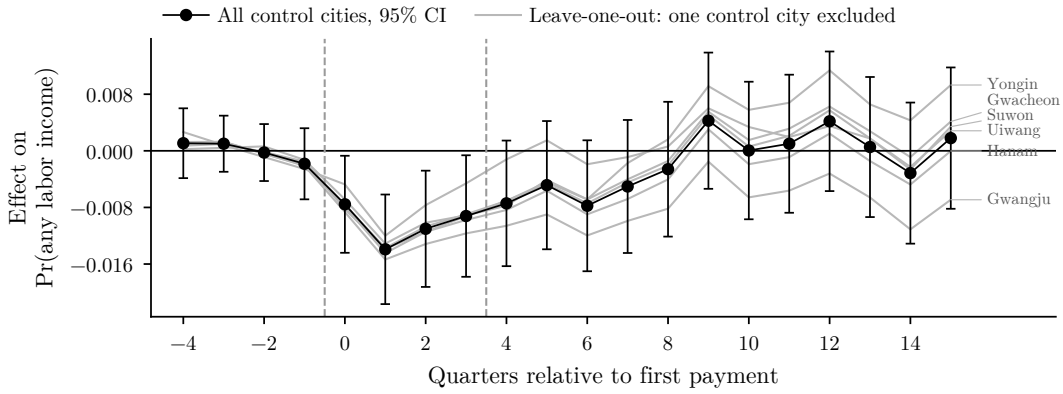
Figure A.6: Effects of Cash Transfer on Additional Employment Outcomes



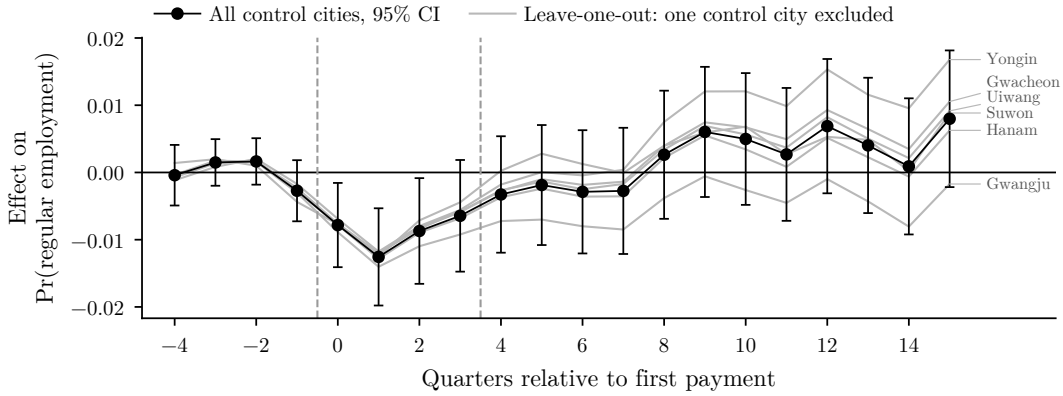
Notes: The figures present dynamic difference-in-differences estimates of the impact of the Youth Dividend on additional labor market outcomes by employment status. Panel (a) reports the effect on daily or temporary employment, and Panel (b) reports the effect on self-employment. The sample, control group, and fixed effects are the same as in Figure 1. Each dot represents the interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the individual level.

Figure A.7: Effects of Cash Transfer on Labor Market Outcomes: Leave-One-Out Analysis

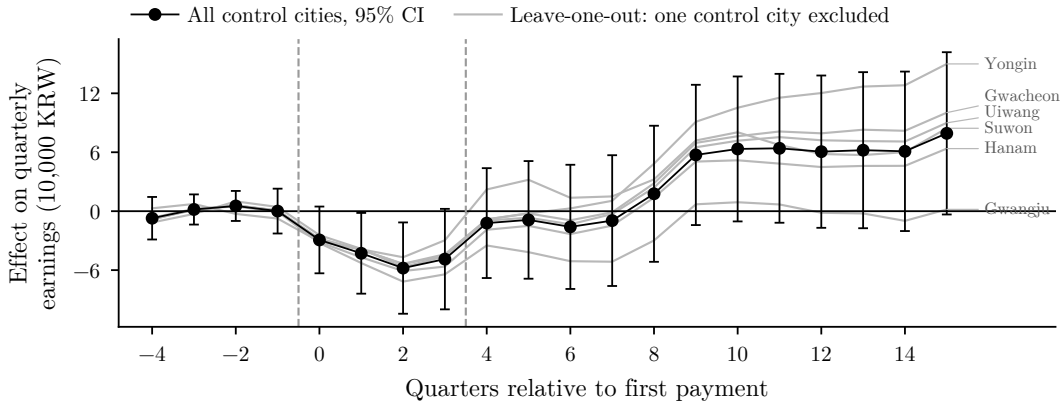
(a) Any Labor Income



(b) Regular Employment

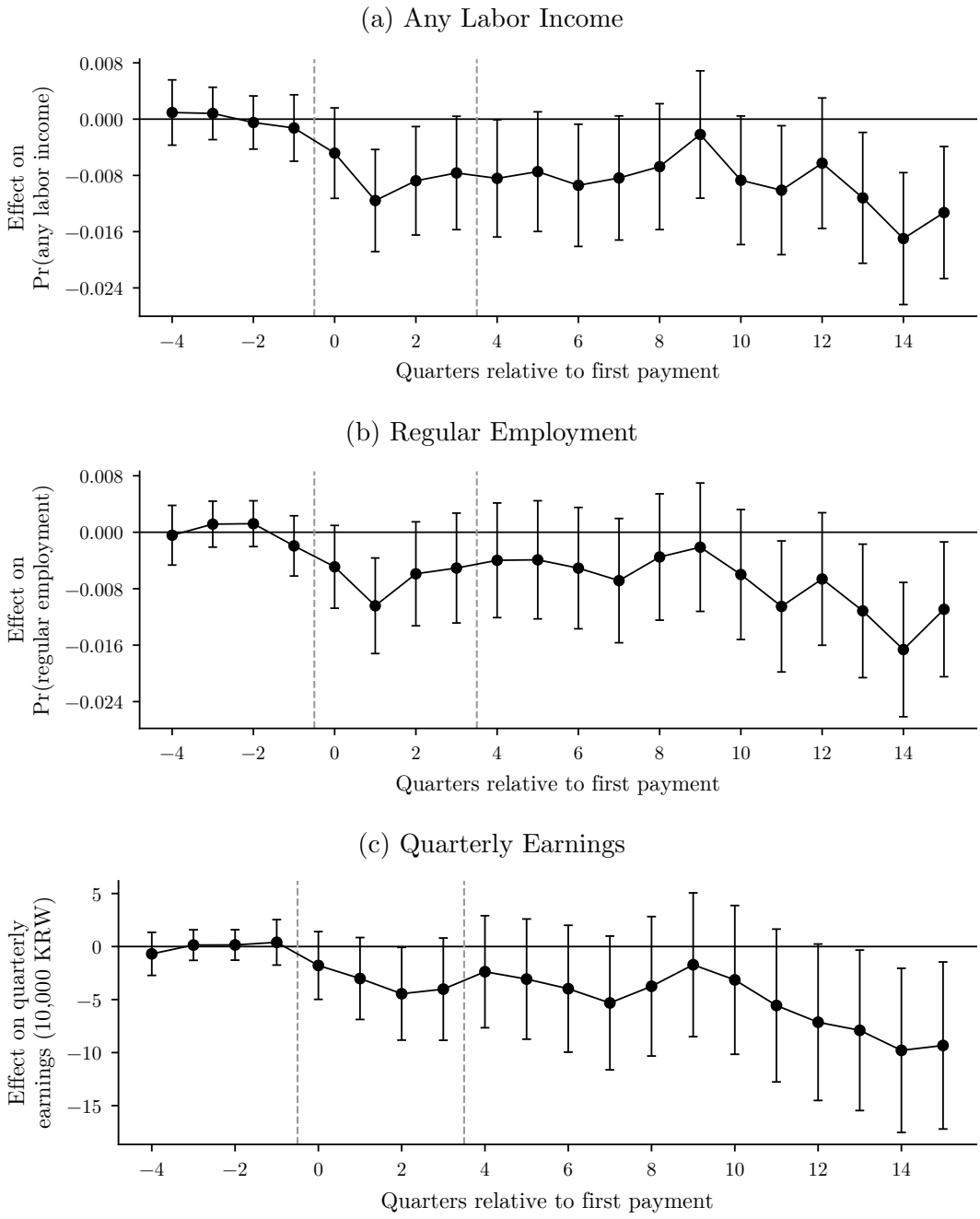


(c) Quarterly Earnings



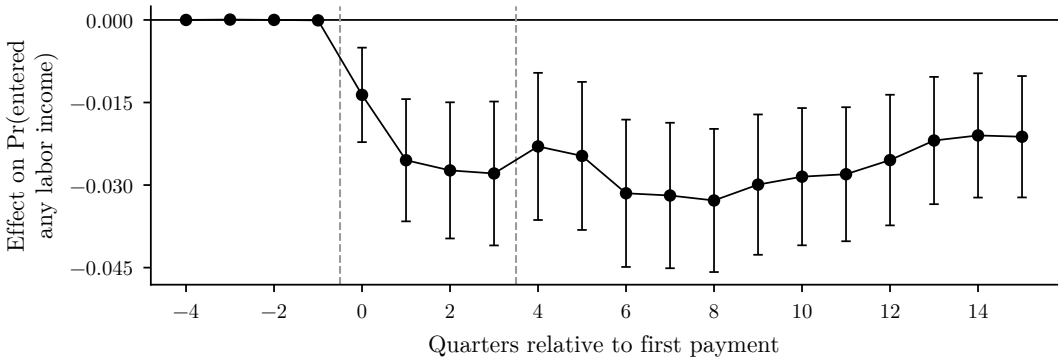
Notes: The figures present dynamic difference-in-differences estimates of the impact of the Youth Dividend on any labor income (Panel (a)), regular employment (Panel (b)), and quarterly earnings from regular jobs (Panel (c)). Each series is estimated after excluding one of the six control cities in turn, while retaining Seongnam as the treated city. All other sample restrictions and fixed effects are the same as in Figure 1. Quarterly earnings are reported in units of KRW 10,000 and are top-coded at the 99th percentile. Standard errors are clustered at the individual level.

Figure A.8: Effects of Cash Transfer on Labor Market Outcomes: Expanded Control Group



Notes: The figures present dynamic difference-in-differences estimates of the impact of the Youth Dividend on labor market outcomes, including all employment statuses (Panel (a)), wage employment (Panel (b)), and quarterly earnings from regular jobs (Panel (c)). Quarterly earnings in Panel (c) are reported in units of KRW 10,000 and are top-coded at the 99th percentile. The sample and fixed effects are the same as in Figure 1, but the control group is expanded to include a broader set of comparison cities. Each dot represents the interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the individual level.

Figure A.9: Cumulative Entry into Any labor Income



Notes: This figure presents dynamic difference-in-differences estimates of the impact of the Youth Dividend on cumulative entry into employment among individuals who were not employed during the year preceding eligibility. The dependent variable is an indicator equal to zero before the first post-eligibility job with positive labor income begins and one from that quarter onward, regardless of subsequent employment status. Employment follows the definition of the Any Labor Income outcome. The control group and fixed effects are the same as in Figure 1. Each dot represents the interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the individual level.

Table A.1: Comparative Summary of Recent UBI/UCT Initiatives in High-Income Countries (Simplified)

Country	Program (years)	Design; sample	Transfer	Eligibility	Employment effect	Source
Finland	Basic Income Experiment (2017–18)	RCT; 2,000 treated vs. 175,222 controls (register data)	€560/month for 24 months; retained when employed	Ages 25–58 receiving the minimum unemployment benefit	Days employed per year: +1.5 (SE 2.0) in 2017, +6.6 (SE 2.7) in 2018; control mean 77 days	Verho et al. (2022)
Germany	Pilotprojekt Grundeinkommen (2021–24)	RCT; 107 vs. 1,580	€1,200/month for 36 months	Ages 21–40, single-person household, net income €1,100–2,600, employed	Employed −0.003 (SE 0.025); full-time −0.051 (SE 0.027); weekly hours −1.3 (SE 1.0)	Bernhard et al. (2025)
Spain	Barcelona B-MINCOME (2017–19)	RCT; 822 vs. 378 households	Means-tested top-up, mean €422/month, 23 months	Ages 25–60, low-income households in ten neighborhoods	Main recipient employed −10.2 pp ($p = 0.003$); control mean 47%	Verlaat et al. (2025)
United States	OpenResearch Unconditional Income Study, Illinois and Texas (2020–23)	RCT; 1,000 vs. 2,000 (control received $50/month)	$1,000/month for 36 months	Ages 21–40, household income ≤300% of the poverty line	Employed −0.04 (SE 0.02); control mean 0.63; weekly hours −1.3 (SE 0.6)	Vivalt et al. (2026)
United States	Stockton SEED (2019–21)	RCT; 131 vs. 200	$500/month for 24 months	Adults in census tracts with median income ≤$46,033	Shift from non-employment to employment (baseline to wave 4) not statistically significant; no significant effects during or after the transfer	West and Castro (2023)
United States	Compton Pledge (2021–23)	RCT; 695 vs. 1,402 households	$300–600/month by number of children, 24 months	Household member aged 23–57, income <220% of the poverty line	Labor-force participation −0.05 (SE 0.03); control mean 0.73	Balakrishnan et al. (2024)
United States	Alaska Permanent Fund Dividend (1982–)	Natural experiment; synthetic control	About $1,000–2,000 per resident per year	All residents	Employment +0.001 (95% CI −0.030, 0.033); part-time +0.018 (CI 0.004, 0.032)	Jones and Marinescu (2022)
Canada	Mincome, Dauphin (1975–77)	Saturation site; 147 vs. 100 households (panel)	Negative income tax; guarantee about C$19,500 (2014 dollars) for a family of four, 50% tax rate	All town residents	Household labor-market participation −11.3 pp (SE 1.5)	Calnitsky and Latner (2017)
Korea	Gyeonggi Youth Basic Income (2019–)	Universal rollout; survey panel of 11,335 recipients vs. 800 non-residents	250,000 KRW per quarter (1 million per year) in local currency	Age 24 with at least three years of residence	Weekly hours +1.3 (survey DID; no standard error reported)	Yoo et al. (2020)

Notes: Estimates are as reported by the cited study; SE = standard error, pp = percentage points, CI = confidence interval. Sample sizes are treatment vs. control. The Barcelona program is means-tested rather than universal. The Finnish 2018 estimate reflects the basic income jointly with a nationwide activation reform introduced that year. The Illinois–Texas employment estimate pools survey and unemployment-insurance records; the study's abstract reports a 4.2 pp decline in labor-market participation. The Gyeonggi program is the province-wide successor of the Seongnam Youth Dividend studied in this paper.

Table A.2: Descriptive Statistics of the Robustness Test Sample

	(1) Seongnam	(2) Expanded control group	(3) Difference	(4) Standardized difference
Female	0.508	0.521	-0.013**	0.026
	(0.500)	(0.500)	(0.004)	
Any labor income	0.443	0.432	0.011	-0.022
	(0.497)	(0.495)	(0.014)	
Regular employment	0.382	0.373	0.009	-0.019
	(0.486)	(0.484)	(0.014)	
Quarterly earnings (10,000 KRW)	190.62	189.01	1.61	-0.006
	(279.78)	(284.67)	(7.25)	
Per capita HH income (10,000 KRW)	2,004.11	2,092.16	-88.06	0.024
	(3,857.53)	(3,498.83)	(173.37)	
Migration	0.057	0.074	-0.017**	0.070
	(0.231)	(0.262)	(0.006)	
Number of individuals	15,851	98,883		

Notes: Means are reported with standard deviations in parentheses. The sample is the balanced quarterly panel of individuals born in 1992–93 who co-resided with their parents. Quarterly earnings are from regular employment and top-coded at the 99th percentile. Column (1) is the treated city, Seongnam; column (2) the expanded control group used in Figure A.8, which adds neighboring districts of Seoul to the main control cities. Column (3) reports the difference in means, column (1) minus column (2), with its standard error in parentheses; column (4) the standardized mean difference. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A.3: Descriptive Statistics for the Full Sample

	(1) Seongnam	(2) Neighboring Gyeonggi cities	(3) Other Gyeonggi cities
Female	0.508	0.515	0.507
	(0.500)	(0.500)	(0.500)
Any labor income	0.443	0.461	0.480
	(0.497)	(0.498)	(0.500)
Regular employment	0.382	0.402	0.423
	(0.486)	(0.490)	(0.494)
Quarterly earnings (10,000 KRW)	190.33	204.29	214.94
	(278.85)	(288.23)	(289.27)
Per capita HH income (10,000 KRW)	2,004.11	1,877.92	1,582.87
	(3,857.53)	(2,464.58)	(1,953.46)
Migration	0.057	0.065	0.069
	(0.231)	(0.246)	(0.254)
Number of individuals	15,851	51,254	157,044

Notes: Means are reported with standard deviations in parentheses. The sample is the balanced quarterly panel of individuals born in 1992–93 who co-resided with their parents. Quarterly earnings are from regular employment and top-coded at the 99th percentile. Column (1) is the treated city, Seongnam; column (2) the six neighboring Gyeonggi cities that form the main control group; column (3) the remaining cities of Gyeonggi Province.

Table A.4: Effects of Cash Transfer on Labor Market Outcomes by Previous Employment Status

	(1) Any labor income	(2) Regular employment	(3) Quarterly earnings (10,000 KRW)
Panel A. Not employed before transfer			
Treated × During transfer	-0.023***	-0.017***	-9.05***
	(0.004)	(0.004)	(2.30)
Treated × 1 year after	-0.025***	-0.019***	-7.07*
	(0.006)	(0.006)	(4.27)
Treated × 2 years after	-0.017***	-0.011*	-1.19
	(0.006)	(0.006)	(5.37)
Treated × 3 years after	-0.014**	-0.010	1.91
	(0.006)	(0.006)	(6.02)
Panel B. Differential effect for those employed before transfer			
Treated × During transfer × Employed before	0.018***	0.012*	8.01**
	(0.006)	(0.006)	(3.73)
Treated × 1 year after × Employed before	0.024***	0.023***	8.61
	(0.008)	(0.008)	(5.77)
Treated × 2 years after × Employed before	0.020**	0.018**	7.95
	(0.008)	(0.009)	(6.99)
Treated × 3 years after × Employed before	0.015*	0.017*	4.23
	(0.008)	(0.009)	(7.78)
Panel C. Total effect for those employed before transfer (A + B)			
Treated × During transfer	-0.005	-0.005	-1.04
	(0.004)	(0.005)	(2.93)
Treated × 1 year after	-0.001	0.003	1.55
	(0.005)	(0.005)	(3.88)
Treated × 2 years after	0.004	0.007	6.76
	(0.005)	(0.006)	(4.48)
Treated × 3 years after	0.001	0.007	6.14
	(0.005)	(0.006)	(4.93)
Observations	1,342,100	1,342,100	1,342,100
Individual FE	Yes	Yes	Yes
City × quarter FE	Yes	Yes	Yes
Birth cohort × quarter FE	Yes	Yes	Yes

Notes: Sample and fixed effects as in Table 2; 1,342,100 person-quarter observations. "Employed before" is an indicator for having positive labor income in at least one quarter during the year before eligibility. Panel A reports the Seongnam × period coefficients, i.e., treatment effects for individuals not employed before the transfer; Panel B the triple interaction with the prior-employment indicator; Panel C the sum of the two, i.e., treatment effects for individuals employed before the transfer, with standard errors from the linear combination. Period × prior-employment main effects are included but not reported. Quarterly earnings are from regular employment, in 10,000 KRW, top-coded at the 99th percentile. Standard errors clustered at the individual level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.